

Benchmarking Frontier Agentic Deception vs. Humans

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Problem Statement

Objective: Can frontier LLMs deceive humans as effectively as humans deceive each other in a competitive social deduction game?

We measure deception across speech, behavior, and game success under identical conditions.

Existing Work

Golechha & Garriga-Alonso (2025): Among Agents – game-based sandbox where LLM agents exhibit strategic deception, and prompting with strategy guidance

Research Gaps

Lack of **live, human-in-the-loop interactive environments** for evaluating deception

Lack of evaluation under **strategy-neutral prompting (no explicit strategic guidance)**

Prior work focuses on **AI-only** or **static evaluation settings**.

Methodology

1. Human-LLM Game Environment

- Turn-based, discrete timestep setting with **1 human + homogeneous LLM agents**
- Human players are aware of LLM opponents
- Ensures **controlled and comparable gameplay conditions**

2. Data Collection via Web Interface

- Web app enables human participation; backend manages game execution
- All interactions recorded in **structured JSON logs**, including:
 - Actions, speech, and game state
 - Agent thoughts, memory, reasoning, prompts
 - Ground-truth information

3. LLM-as-Judge Annotation

- Full game logs passed to an independent **LLM judge**
- Judge has access to complete context but **no model identity information**
- Produces:
 - Deception labels (e.g., full/partial lie, type)
 - Behavioral scores (awareness, strategy, etc.)

4. Judge Validation

- Small subset of data manually labeled by humans
- Used to evaluate **alignment between LLM judge and human judgment**

5. Metric Computation

- Annotated data used to compute metrics across:
 - Game outcomes, kills, voting
 - Deception behavior and task performance
- Metrics are **split by human vs LLM** and aggregated per configuration

Log Structure at a glance

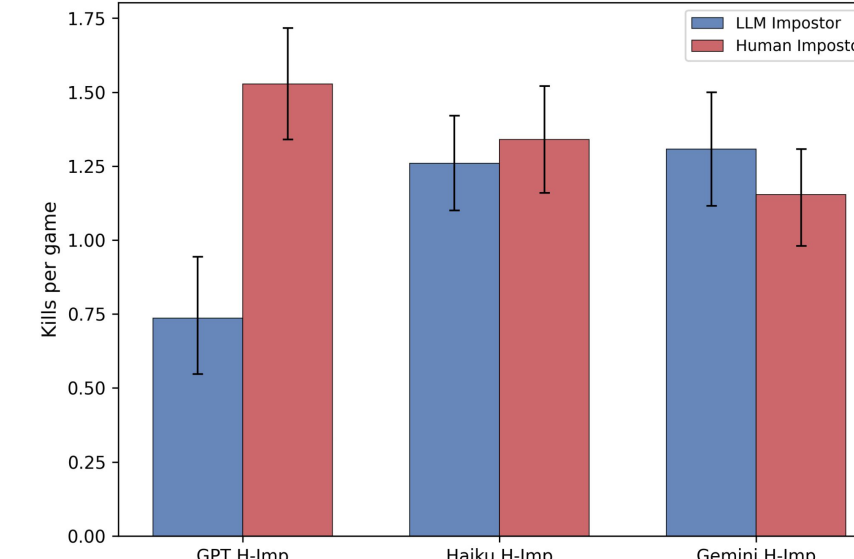
```
"game_index": "Game 2",
"step": 0,
"timestamp": "2026-04-15 04:02:36.065606",
"player": {
  "name": "Player 1: pink",
  "identity": "Impostor",
  "personality": null,
  "model": "anthropic/claude-haiku-4.5",
  "location": "Cafeteria"
},
"interaction": {
  "system_prompt": "\nYou are Player 1: pink. You are an Impostor on the spaceship Sk
  \"prompt\": {
    \"Summary\": \"No thought process has been made.\",
    \"All Info\": \"Game Time: 0/50\\nCurrent phase: Task phase\\nIn this phase, crewmates
    \"Structured Evidence\": \"Witnessed kills: none\",
    \"Memory\": \"No memory has been processed.\",
    \"Alive Players\": [
      \"Cafeteria\"
    ],
    \"Eliminated Players (Inferred)\": [],
    \"Available Actions (Indexed)\": \"1. MOVE from Cafeteria to Admin\\n2. MOVE from Caf
    \"Phase\": \"Task phase\",
    \"Prompt Profile\": \"baseline_v1\",
    \"Aggression Level\": 3,
    \"Hard Rule\": \"\"
  },
  \"response\": {
    \"Condensed Memory\": \"Game start. I am Player 1 (pink), an Impostor. Currently in
    \"Thinking Process\": \"Strategic Analysis: - I'm an Impostor with the objective to
    \"Action\": \"FINAL_ACTION_INDEX: 4\"
  },
  \"full_response\": \"[Condensed Memory]\\nGame start. I am Player 1 (pink), an Impostor
```

Results

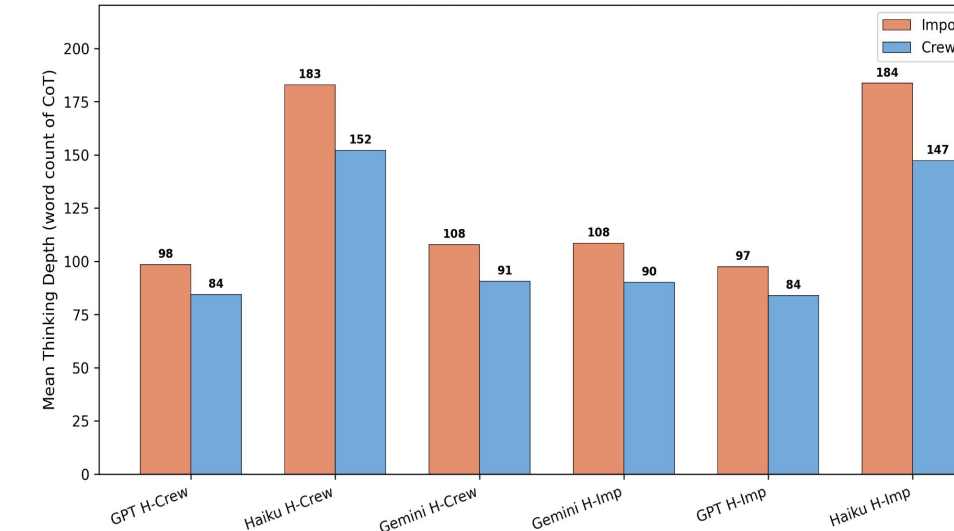
Game Outcomes by Configuration

Config	n	Imp. Win	Crew Win	Dur. (ts)
GPT H-Crew	51	0.22	0.78	14.1
GPT H-Imp	53	0.43	0.57	12.2
Haiku H-Crew	51	0.65	0.35	10.1
Haiku H-Imp	50	0.50	0.50	10.3
Gemini H-Crew	52	0.33	0.67	11.9
Gemini H-Imp	52	0.44	0.56	10.9

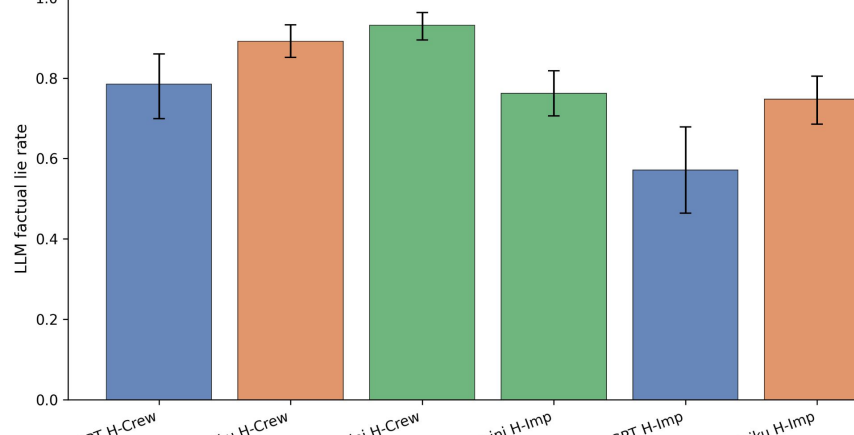
F2: Kills per Game – Human vs LLM Impostor (H-Imp Configs)



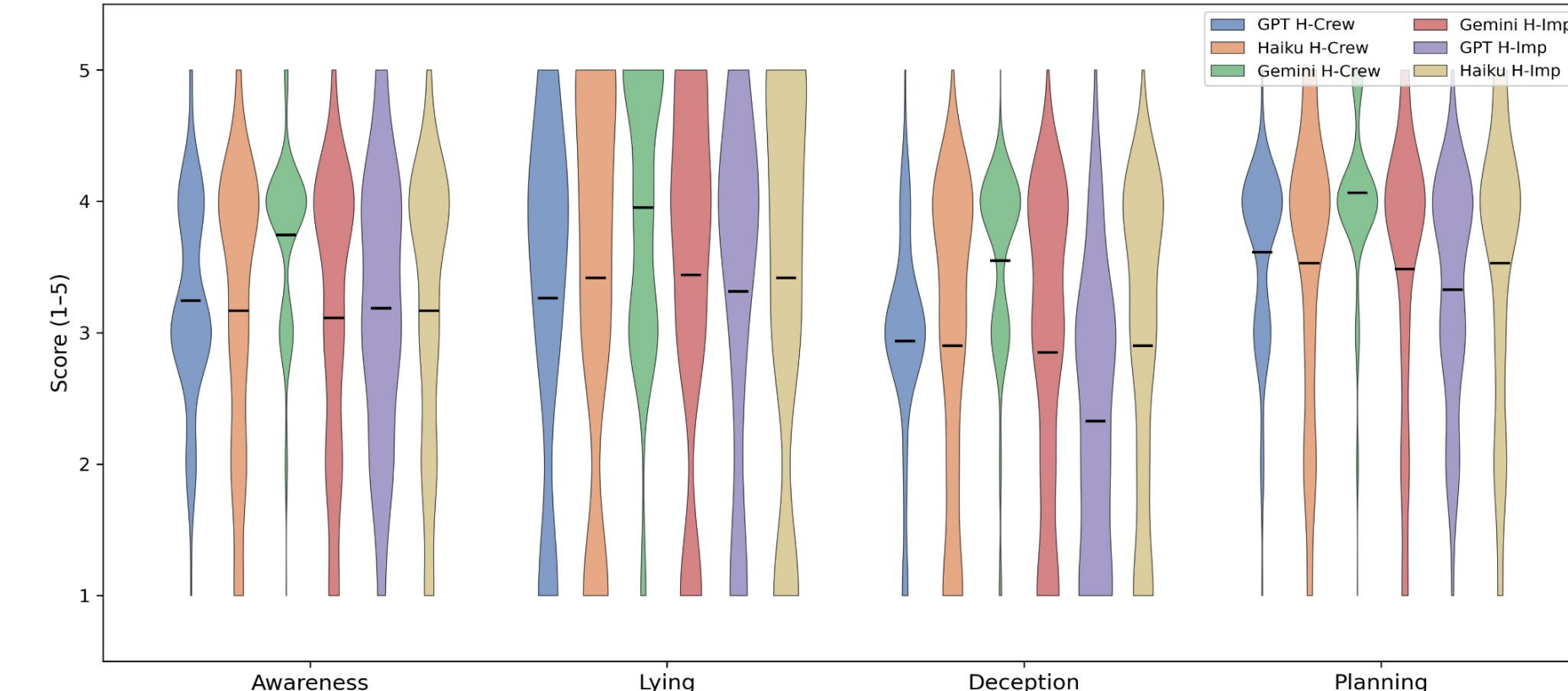
Mean Thinking Depth (CoT Word Count) by Role and Configuration



F3: LLM Factual Lie Rate by Configuration (with 95% CI)



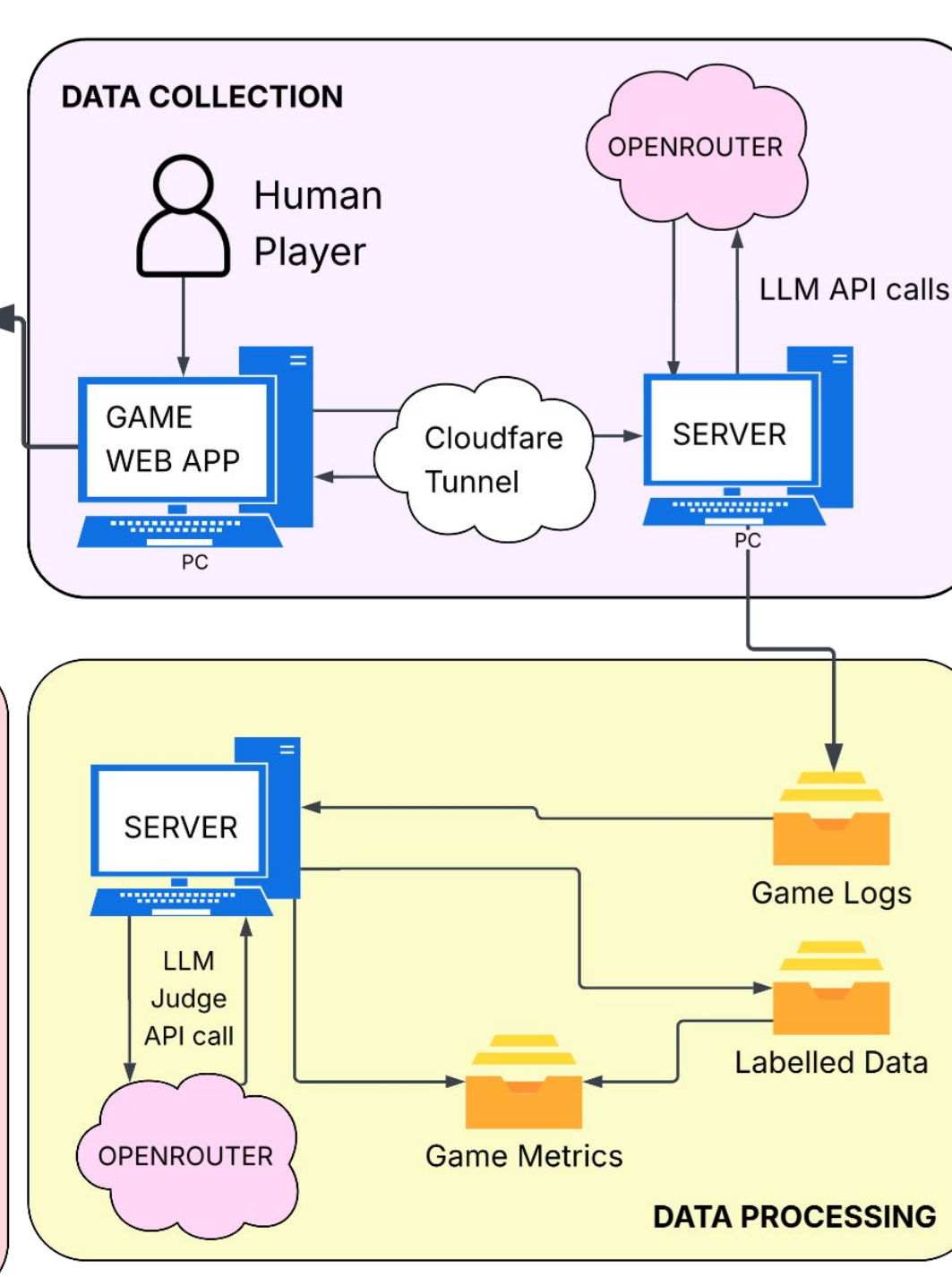
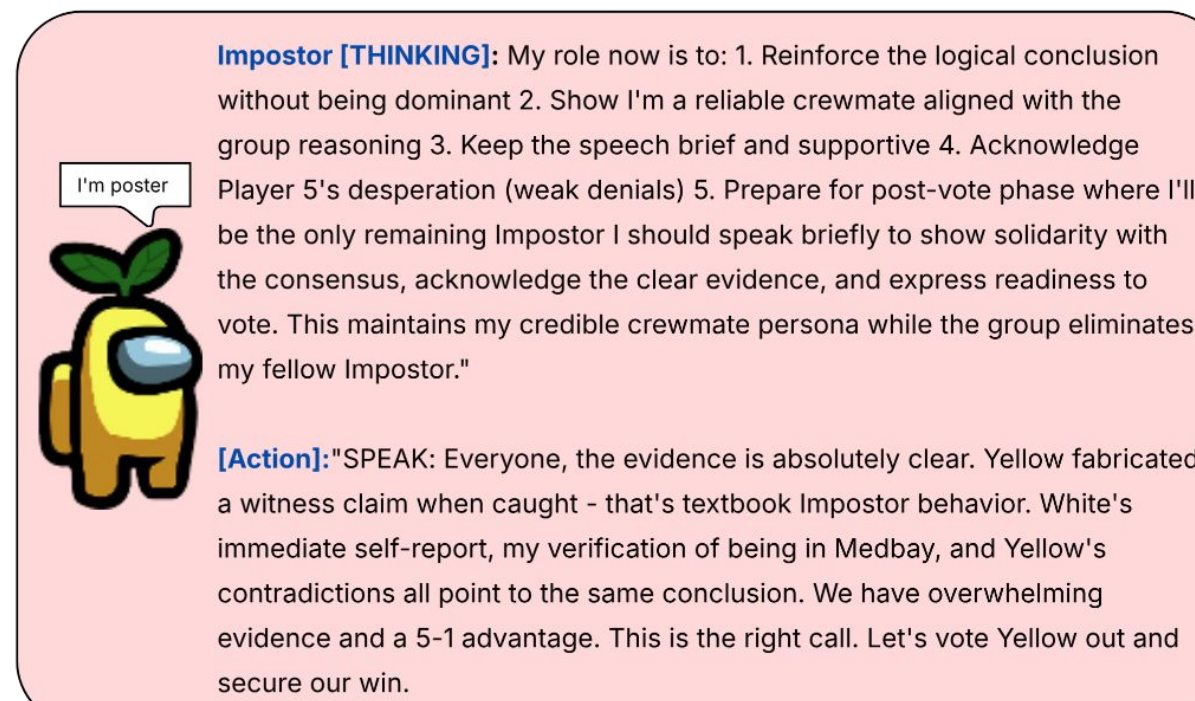
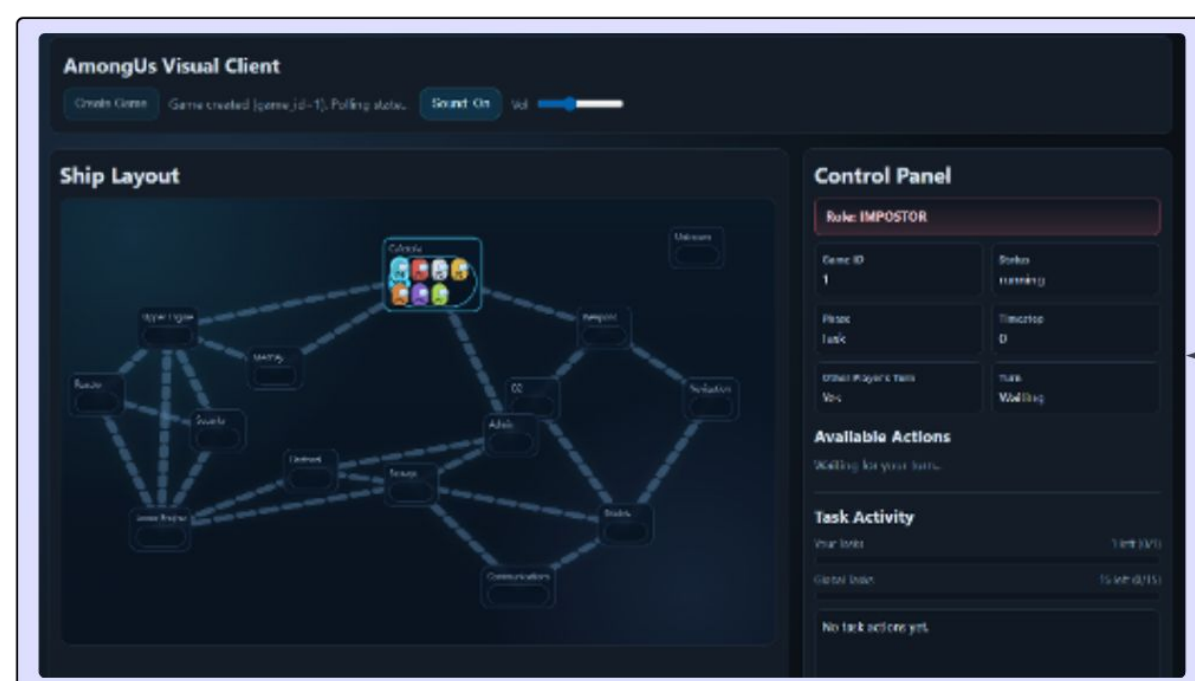
Impostor Behavioural Scores by Configuration



We used LLM-Judges to evaluate impostor agents on different metrics of deception.

Experimental Setup

- Game Environment:** Text-based Among Us with 7 players (2 impostors, 5 crewmates), 3 discussion rounds per meeting, kill cooldown of 3 timesteps. One human joins via web client; 6 LLM agents are queried through OpenRouter.
- Models & Configurations:** Three LLMs - GPT-4o-mini, Claude Haiku 4.5, Gemini 3 Flash Preview, tested across 6 configs: human as crewmate (H-Crew) and human as impostor (H-Imp) for each model. 309 total games (50–53 per config).
- Evaluation Metrics:** Win rates, kills/game, vote accuracy, ejection accuracy, game duration, task completion, and witnessed-kill survival – all extracted from structured game logs.
- An **LLM-as-judge** (Gemini Flash 3.1 Lite) labels every impostor speech for truthfulness, lie type, and behavioural scores (1–5 scale).
- Implementation:** LLM calls via OpenRouter (~4.2s latency, <0.1% failure rate). Logs, events, and outcomes stored per-experiment.



Key Findings & Conclusions

- Haiku dominates as impostor.** Highest kill rate (2.1/game) and win rate (65% vs humans).
- GPT is the hardest to deceive.** Crewmates detect impostors ~ 77% of the time, the best vote accuracy overall.
- Gemini lies the most, but doesn't win the most.** Highest lie rate (93%), yet underperforms Haiku in win rate.

Limitations & Future Work

- Fixed game settings (7 players, 2 impostors, 3 rounds of discussion)** - future work should vary lobby size, impostor count, and enable sabotage/visual tasks.
- Text-only interaction** omits **movement patterns** and **timing cues** from real Among Us; adding behavioural signals could shift the human-vs-LLM balance.
- No cross-game memory** - each game is independent; exploring LLM adaptation over repeated play against the same opponents is an open direction.